# Fake News Detection Using Probabilistic Modeling and Naïve Bayes Inference

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## Abstract

This project investigates automatic fake news detection using probabilistic models, focusing on a Naïve Bayes classifier trained on the ISOT Fake News Dataset. We examine data preprocessing, feature extraction, model inference, hyperparameter tuning, discriminative feature analysis, and temporal generalization. Our findings highlight both the strengths and limitations of probabilistic text classifiers in misinformation detection [1].

## 1 Introduction

Fake news detection is the task of automatically identifying whether a news article is fake or real. The rapid spread of misinformation has serious consequences for public opinion, civic decision making, and trust in democratic institutions. Prior work has shown that false information propagates faster and more broadly than factual reporting [1]. In this project, we study the problem of predicting: $P(\textbf{label} \mid \textbf{text})$ where the label represents either `fake` or `real`, using probabilistic modeling techniques. This problem is closely connected to probabilistic reasoning because linguistic patterns, stylistic markers, and topical cues in misinformation can be modeled as random variables with measurable statistical regularities. Naïve Bayes has been widely used for text classification due to its simplicity, interpretability, and strong empirical performance on bag-of-words features. Studying fake news through a probabilistic lens therefore helps us understand not only how to classify misinformation but also the structure of the cues that distinguish it.

## 2 Dataset

We use the ISOT Fake News Dataset[2] , which contains 44,898 news articles labeled as `fake` or `real`. Of these, 23,481 are fake and 21,417 are real. Each entry includes a title, full article text, subject category, and publication date. The real articles were collected from Reuters.com, while the fake articles come from websites identified by PolitiFact and Wikipedia as unreliable sources. The dataset covers primarily political and world news content, providing linguistic variation that supports effective probabilistic modeling of fake versus real reporting.

### 2.1 Exploratory Data Analysis (EDA)

The histograms 1 compare character and word counts across fake and real articles, showing that both distributions are similarly skewed with fake articles tending to be slightly longer on average. The subject distribution 2 shows that most articles belong to politics and world news categories, reflecting the dataset's strong focus on politically relevant content. The top-word comparison 3 shows distinct lexical patterns, with fake articles emphasizing political figures and stance-heavy terms while real articles feature more neutral reporting words.

[1] Code is avalaible here - `https://github.com/Raghav10j/CSE250`

[2] `https://www.kaggle.com/datasets/csmalarkodi/isot-fake-news-dataset`

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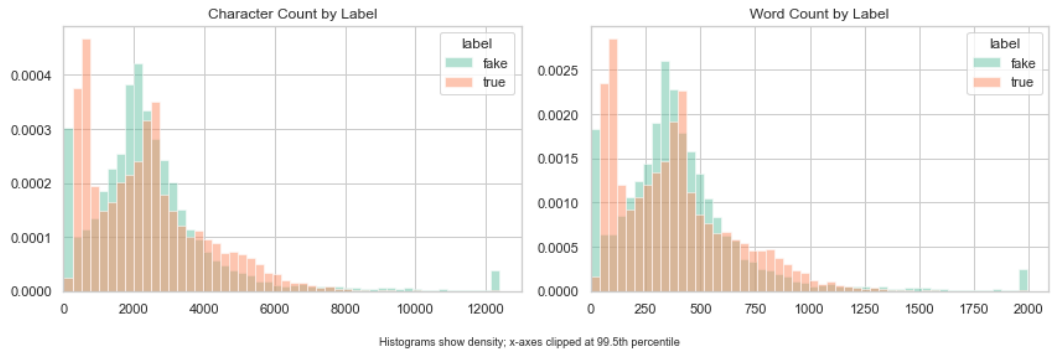



Figure 1: Character and word count distributions for fake and real news articles.

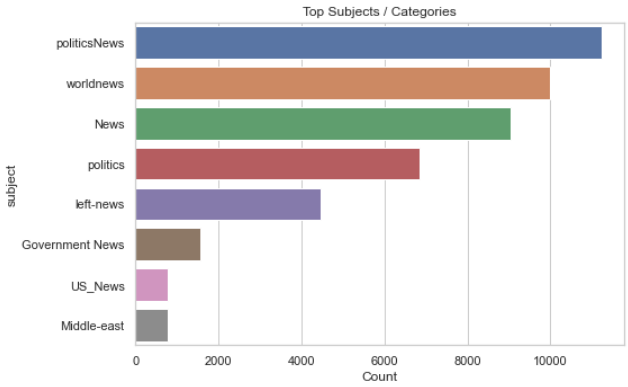



Figure 2: Distribution of article subjects.

### 2.2 Data Preprocessing

Before modeling, we applied the following preprocessing steps to ensure clean and consistent input data - (1) The `title` and `text` fields were concatenated into a single `combined_text` field to provide a richer representation of each article. (2) Entries with missing or empty text were removed, and all labels were encoded numerically, with `fake` mapped to 0 and `true` mapped to 1. (3) Duplicate articles were also eliminated to prevent data leakage. (4) Finally, the dataset was split into training and test sets using an 80–20 ratio with stratification by label to preserve the original fake-to-real class balance.

## 3 Problem Formulation, Modeling, and Inference

### 3.1 Problem Statement

Given a dataset $D = \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$ of $n = 44{,}898$ news articles, the goal is to build a classifier that predicts whether each article is fake or real. Each input $x_i \in X$ is a text document and each label $y_i \in Y = \{0, 1\}$, where 0 denotes `fake` and 1 denotes `true`. The objective is to learn a function $f : X \to Y$ that assigns the most probable class label to a given news article based on its textual content.

### 3.2 Mathematical Formulation

Let $X = \{x_1, \ldots, x_n\}$ denote the documents, $Y = \{0, 1\}$ denote the class labels, and $P(Y \mid X)$ denote the posterior probability of the class given a document. The classification objective is: $\hat{y} = \arg\max_{y \in Y} P(Y = y \mid X = x)$ Using Bayes' theorem:

$$P(Y = y \mid X = x) = \frac{P(X = x \mid Y = y)P(Y = y)}{P(X = x)}.$$

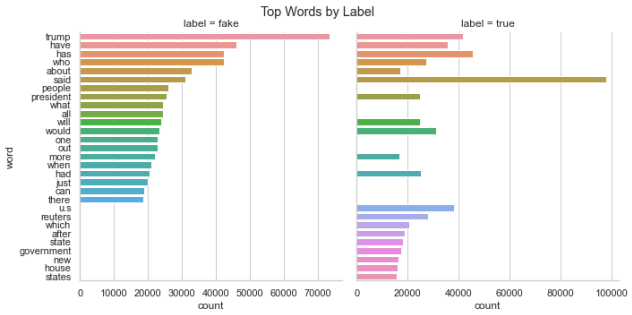



Figure 3: Top frequent words in fake versus real articles

Since $P(X = x)$ is constant across classes: $\hat{y} = \arg\max_{y \in Y} P(X \mid Y = y)\, P(Y = y)$. This formulation treats fake news detection as a probabilistic inference task in which the goal is to model how likely the observed text is under each class.

### 3.3 Probabilistic Model: Multinomial Naïve Bayes

We use the Multinomial Naïve Bayes classifier, which corresponds to a simple Bayesian network with the class variable $Y$ as the parent node and all word occurrences in the document as conditionally independent children.
**Model Assumptions:** (1) Words in a document are conditionally independent given the class label. (2) Documents are generated from a multinomial distribution over vocabulary terms. (3) Class priors reflect empirical frequencies in the training set.

For a document $x = (w_1, \ldots, w_m)$: $P(X = x \mid Y = y) \approx \prod_{i=1}^{m} P(w_i \mid Y = y)$.

Under a bag-of-words count representation: $P(X = x \mid Y = y) \propto \prod_i P(w_i \mid Y = y)^{\text{count}(w_i)}$.

The log-likelihood form is: $\log P(Y = y \mid X = x) = \log P(Y = y) + \sum_i \text{count}(w_i) \log P(w_i \mid Y = y) + C$.

#### Parameter Estimation

Model parameters are estimated with Maximum Likelihood Estimation:

$$P(Y = y) = \frac{\text{number of documents with class } y}{n},$$

$$P(w_i \mid Y = y) = \frac{\text{count}(w_i, y) + \alpha}{\text{total word count in class } y + \alpha|V|}.$$

Here $\alpha$ is the Laplace smoothing parameter and $|V|$ is the vocabulary size.

#### Inference Rule

For a new document $x$:

$$\hat{y} = \arg \max_{y \in \{0,1\}} \left[ \log P(Y = y) + \sum_i \text{count}(w_i) \log P(w_i \mid Y = y) \right].$$

Exact inference is feasible because all required probabilities can be computed directly from observed counts.

### 3.4 Text Representation: TF-IDF Vectorization

To apply the Multinomial Naïve Bayes model, documents must be converted to numeric feature vectors.
**TF-IDF** Each document is represented using TF-IDF weights: $\text{TF-IDF}(w, d) = \text{TF}(w, d) \cdot$

$\log\left(\frac{N}{df(w)}\right)$, where $\text{TF}(w, d)$ is the term frequency and $df(w)$ is the number of documents containing word $w$.

**N-gram Features** We use bigrams (ngram_range $= (2, 2)$) to capture short contextual patterns such as "fake news" and "president said."

**Feature Selection** To reduce dimensionality and improve generalization: (1) `max_features` = 25,000, (2) `min_df` = 2, (3) `max_df` = 0.95, (4) English stop words removed. These preprocessing steps align the feature space with the assumptions of the multinomial model and support effective inference.

### 3.5 Experimental Setup

**Implementation Details -** All preprocessing, exploratory analysis, and visualizations were implemented in Python using `pandas`, `numpy`, `matplotlib`, and `seaborn`. Text vectorization, model training, and evaluation relied on the `scikit-learn` library, where we used its implementations of `Multinomial Naïve Bayes`.

**Train–Test Split -** We used an 80–20 stratified split to preserve the real/fake class balance. All hyperparameter tuning was performed only on the training set to avoid leakage.

**Hyperparameter Tuning -** We tuned the parameters most influential for Naïve Bayes text classification: (1) Smoothing parameter - tested values in 0.01, 0.1, 0.5, 1.0 to analyze over-smoothing vs. under-smoothing effects, (2) N-gram range - compared unigrams, bigrams, and unigram–bigram combinations, and (3) Vocabulary size - evaluated values between 5,000–30,000 to balance expressiveness and computational cost.

**Model Comparison -** We compared above discussed Multinomial Naïve Bayes with following two models. (1) *Complement Naïve Bayes + TF-IDF* - Designed to handle imbalanced datasets more robustly. (2) *Multinomial Naïve Bayes + CountVectorizer* - Included to isolate the effect of TF-IDF reweighting vs. raw counts.

**Evaluation Metrics -** We evaluate model performance using standard binary classification metrics. We report accuracy, precision, recall, F1-score, and ROC-AUC.

## 4 Result and Discussion

### 4.1 Hyperparameter Tuning

For the Multinomial Naïve Bayes model, the primary hyperparameter is the smoothing constant $\alpha$, which controls how unseen words are treated and prevents zero probabilities in the likelihood estimation. Performance peaks at $\alpha = 0.1$ and decreases as smoothing increases 4. We tuned the

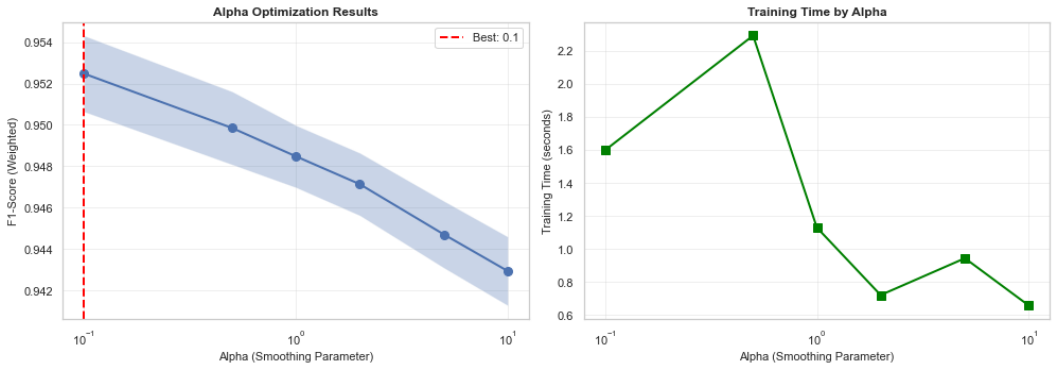



Figure 4: F1-scores and training times for different values of the smoothing parameter $\alpha$.

$n$-gram range by comparing unigrams, unigram–bigram combinations, and bigrams alone, and we skipped trigrams due to their high computational cost. The bigram–bigram model provides the highest F1-score while keeping the vocabulary size reasonable 5. We tuned the `max_features` parameter to balance vocabulary size and predictive performance, noting that F1-scores improved with larger vocabularies and peaked near 25,000 features.

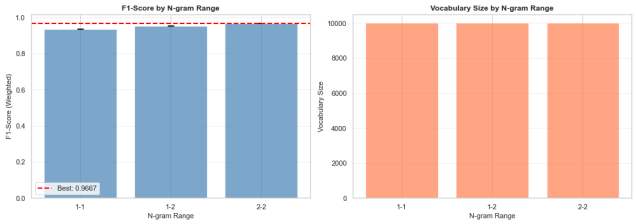



Figure 5: F1-scores and vocabulary sizes for different $n$-gram ranges.

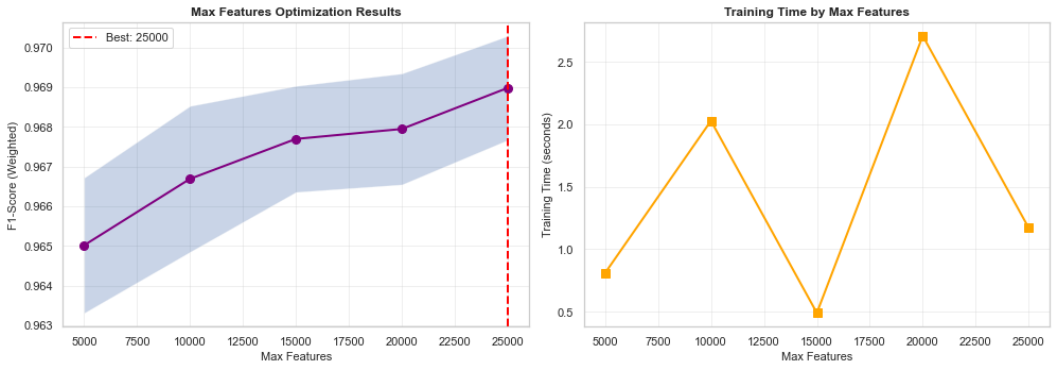



Figure 6: F1-scores and training times for different values of `max_features`.

## 4.2 Main Results

The tuned Multinomial Naïve Bayes model achieved strong performance on the test set, with an accuracy of 0.9482 and an F1-score of 0.9482, indicating balanced precision and recall. The precision of 0.9483 shows that most predicted real articles were correct, and the ROC AUC of 0.9877 reflects excellent separability between fake and real news. These results demonstrate that the optimized model generalizes well and effectively distinguishes between the two classes. The confusion matrices 7 show that the optimized model performs consistently well on both the training and test sets, with very low misclassification rates. Most fake and real articles are correctly identified, and the small number of false positives and false negatives indicates strong generalization without overfitting. The balanced performance across classes further supports the high F1-score observed earlier. The ROC curve 8a shows that the optimized model achieves excellent discrimination between fake and real news, with the curve staying near the top-left corner. The resulting AUC of 0.9943 indicates that the classifier almost perfectly separates the two classes, far outperforming a random baseline. This further confirms the robustness and reliability of the tuned Naïve Bayes model. The log-odds analysis 8b highlights clear lexical differences between fake and real news. Fake articles often contain bigrams related to images, screenshots, and alternative media sources, while real articles prominently feature location and agency-specific bigrams tied to Reuters reporting. These patterns show that stylistic and source cues are strong indicators for classification.

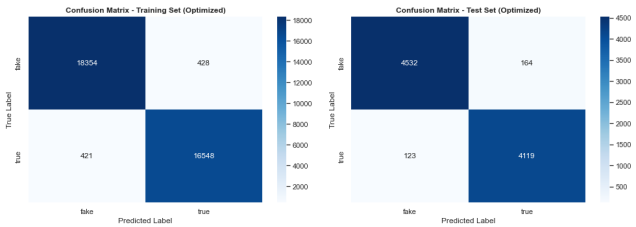
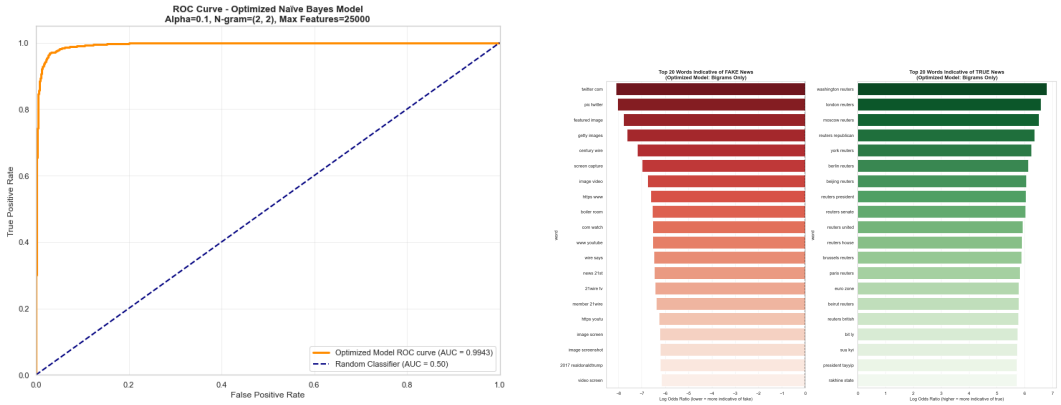



Figure 7: Confusion matrices for Multinomial Naïve Bayes model on the training and test sets




(a) ROC curve for the optimized Naïve Bayes model (b) Top bigrams indicative of fake vs. real news

Figure 8: ROC Curve and feature importance

### 4.3 Model Comparison

We compared three optimized models: MultinomialNB with TF-IDF, ComplementNB with TF-IDF, and MultinomialNB with CountVectorizer. The results show that the TF-IDF MultinomialNB model achieves the best overall performance, with an accuracy of about 97 percent, an F1-score near 0.97, and a ROC AUC close to 0.99. ComplementNB performs slightly worse across all metrics, and the CountVectorizer-based MultinomialNB model trails further behind due to its simpler representation. Overall, the TF-IDF representation combined with MultinomialNB provides the most effective balance of accuracy, robustness, and interpretability.

### 4.4 Learned Probability Distributions

The Naïve Bayes model learns separate conditional probability distributions for each word in the vocabulary, capturing how strongly a term is associated with fake or true news. The distribution of log-likelihood ratios shows that most words cluster near zero, with a long tail of highly discriminative terms. The scatter plot 10 comparing $P(w \mid \text{fake})$ and $P(w \mid \text{true})$ reveals clear separation between words that favor each class, indicating that the model captures meaningful lexical patterns that support accurate classification.

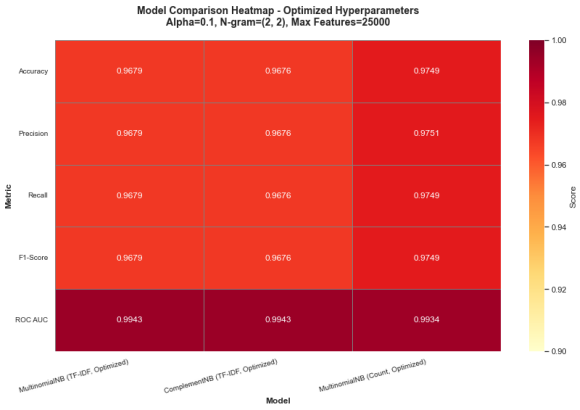



Figure 9: Heatmap comparing MultinomialNB (TF-IDF), ComplementNB (TF-IDF), and MultinomialNB (Count) models

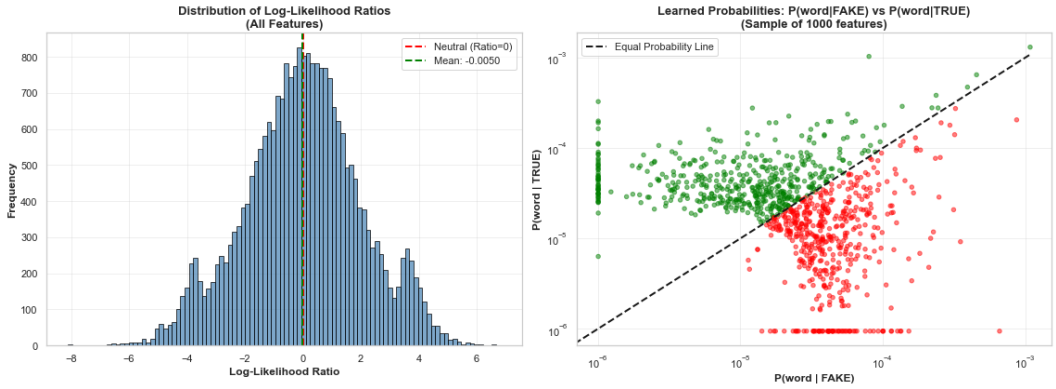



Figure 10: Distribution of log-likelihood ratios and scatter plot of learned conditional probabilities $P(w \mid \text{fake})$ vs. $P(w \mid \text{true})$.

### 4.5 Temporal Validation

To assess temporal robustness, we trained the model on articles from 2016 and tested on articles from 2017. Performance dropped from 96.79 percent accuracy under a random split to 91.95 percent with the temporal split, indicating moderate temporal drift. The metrics (F1 = 0.9206, AUC = 0.9853) show that while the model still generalizes well, language and topic shifts over time reduce accuracy compared to i.i.d. evaluation.

## 5 Conclusion

This project demonstrated that a tuned Multinomial Na"ive Bayes model with TF-IDF features can effectively detect fake news, achieving strong accuracy, F1-score, and AUC while revealing clear lexical differences between fake and real articles. Although the model performs well, it is limited by the Na"ive independence assumption, sensitivity to temporal drift, and reliance on a single dataset. Future work could explore ensemble learning methods and more diverse, time-sensitive datasets to improve robustness and generalization.

## 6 Reflections & Contributions

**Raghav Jain**: Raghav contributed to drafting, brainstorming, and coding. He reflected that the project helped connect theoretical concepts introduced by the TA and instructors to real-world implementation, and expressed interest in exploring continual learning approaches for evolving misinformation in future work.
**Ishaan Lonial**: Ishaan contributed to brainstorming, draft polishing, and code setup. Through this project, he deepened his understanding of the bias–variance tradeoff in smoothing and vocabulary selection, and found it especially interesting to see how specific stance-heavy terms emerged through discriminative feature analysis.
**Vincent McCloskey**: Vincent contributed to brainstorming, formalizing results, and preparing the final paper. He reflected that this project strengthened his conceptual understanding of probabilistic models by applying them directly rather than only computing equations, and emphasized how EDA provided essential insight into how the data should be handled.
**Lillian Tran**: Lillian contributed to brainstorming, formalizing results, and final paper preparation. She appreciated the transparency of the Na"ive Bayes model and how feature-level interpretation was straightforward, while also noting its sensitivity to temporal drift due to changing news patterns.
**Kevin Benavente**: Kevin contributed to brainstorming, formalizing results, and preparing the final paper. He noted that the project improved his understanding of probabilistic modeling, TF-IDF interactions with Na"ive Bayes assumptions, and the practical challenges of optimizing large text pipelines and interpreting log-odds for lexical analysis.

**Use of Generative AI:** We used AI assistance to help reformat text, debug code, and generate LaTeX structure, but all modeling, analysis, and empirical results were completed by the team.

## 7 References

## References

[1] Soroush Vosoughi, Deb Roy, and Sinan Aral. The spread of true and false news online. *science*, 359(6380):1146–1151, 2018.